

Predicting Loss Curve of LLM Pretraining

Task 2 Final Project

Group: Tianze Lu · Ruoyu Wang · Feiyue Ye

Student IDs: confirm before submission

GitHub repository: <https://github.com/Xiaoxiao-2002/TDLT2025-final-project.git>

Why this problem matters

- Pretraining loss curves are the most direct signal of whether an LLM training run is healthy.
- If we can predict the loss curve under a new learning-rate schedule, we can reduce schedule tuning cost.
- Task 2 asks us to fit on cosine and predict on WSD, then go beyond the reproduced baselines.

The core question is not just fitting one curve well, but transferring the fitted law across different learning-rate schedules.

Problem setup

- Train on the cosine schedule.
- Predict on WSD, with 8-1-1 as an extra cross-schedule test.
- Use the observed loss curve together with the schedule signal.

The task is to transfer a fitted law across schedules, not only to fit one curve well.

Evaluation

- Main metrics: log-space R^2 and MAPE.
- Secondary metrics: MAE, RMSE, PredE, and WorstE.
- The final comparison focuses on cross-schedule test error.
- Reported numbers use the saved best runs under `experiments/`.

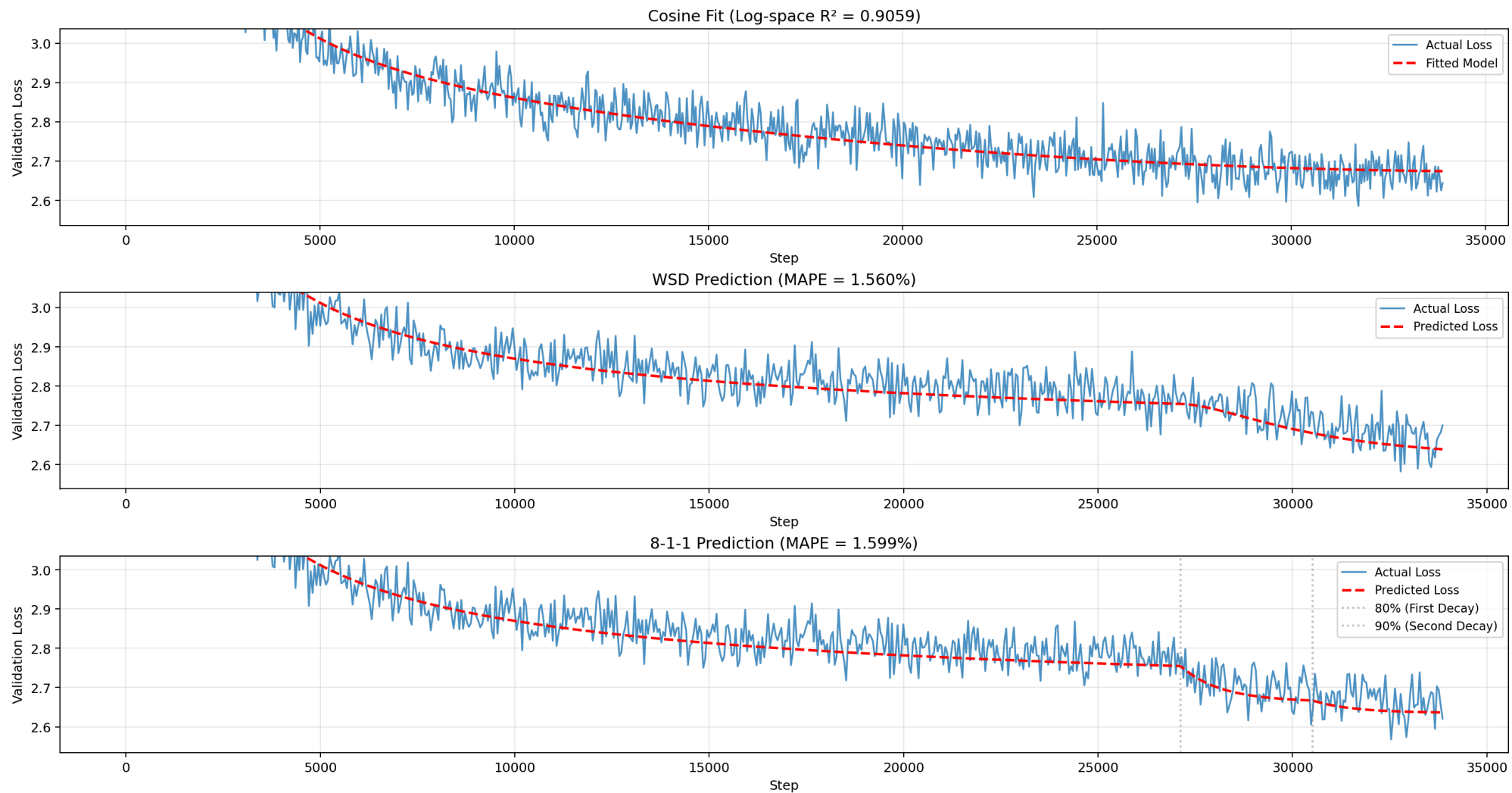
Baseline 1: Tissue / Momentum Law

The simplified form is:

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t)$$

- Fits the cosine schedule reasonably well.
- But it tends to **overpredict early-stage loss**.
- That early transient mismatch is the main weakness we observed.
- Full curve visualization is on the next slide.

Baseline 1: Full Figure



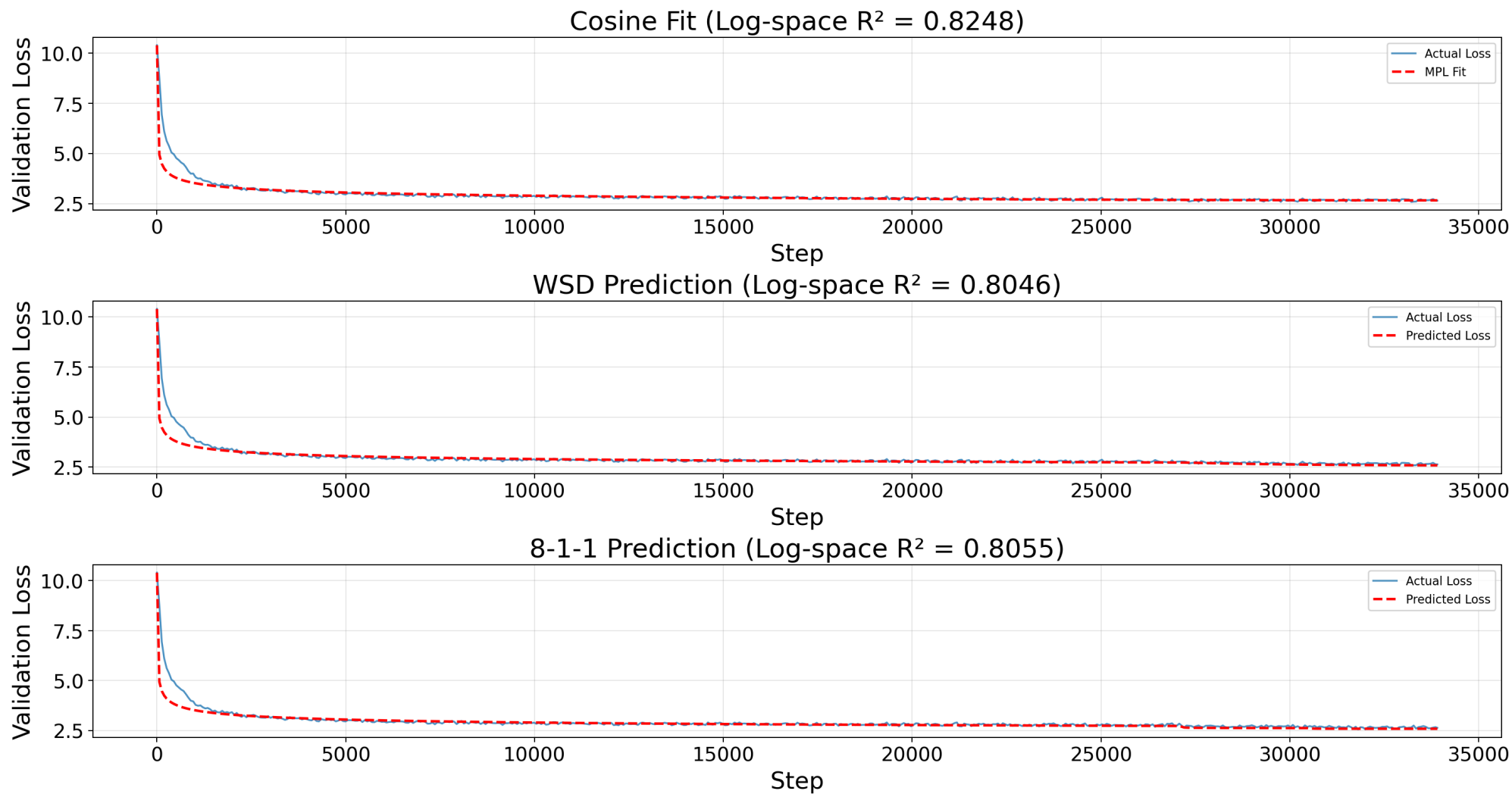
Baseline 2: Multi-Power Law

The 7-parameter MPL is more expressive:

$$L(t) = L_0 + AS_1(t)^{-\alpha} + B \cdot \text{drop}(t)$$

- More flexible seven-parameter form.
- Uses the same course pkl data as the Tissue reproduction.
- Fits on cosine and predicts WSD plus 8-1-1.
- This is the more expressive reproduced baseline.
- Full curve visualization is on the next slide.

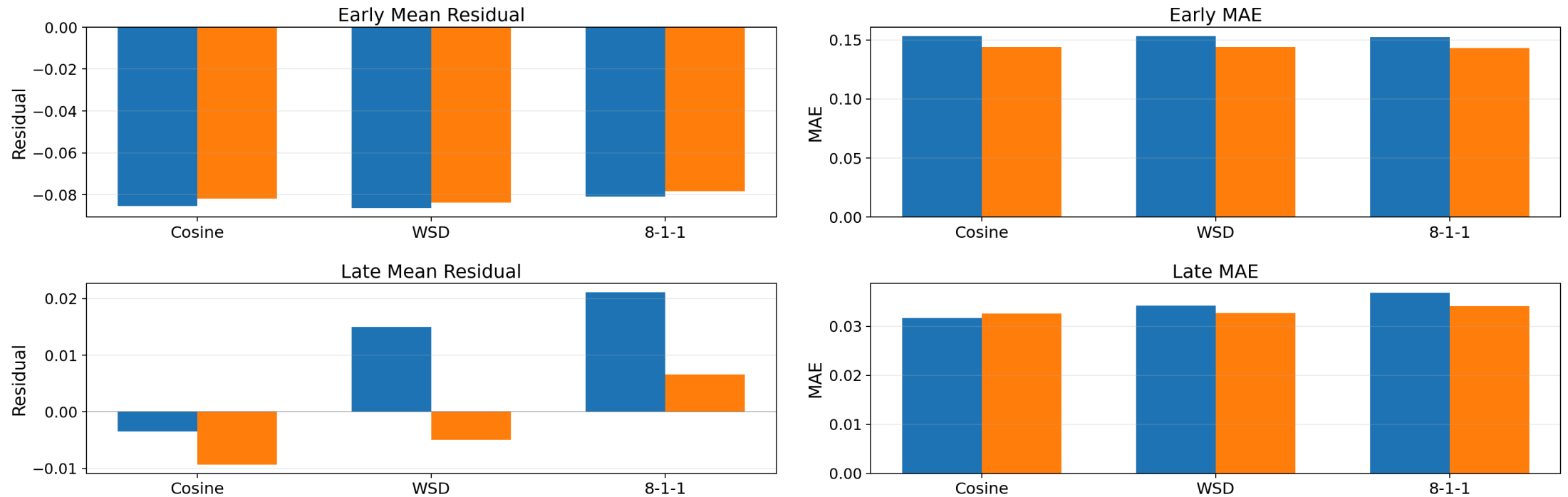
Baseline 2: Full Figure



Failure mode we found

Residual Pattern Diagnosis for Tissue Baseline vs Improved Variant

Blue: Tissue baseline | Orange: Tissue + progress correction



The early-step residuals are consistently negative in the plain Tissue baseline, meaning the model overpredicts the early transient.

Our method candidate

We keep the original Tissue structure and add a small progress-aware correction:

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t) - De^{-\tau p(t)}$$

where $p(t)$ is normalized progress from 0 to 1.

Why this helps

- It changes only the part that fails most clearly.
- It stays simple and interpretable.
- It reduces the early-step mismatch without redesigning the whole model.

Main results

Method	Fit metric	Test metric
Tissue baseline	Cosine log-space $R^2 = 0.905891$	WSD MAPE = 1.5601%; 8-1-1 MAPE = 1.5994%
Tissue + progress correction	Cosine log-space $R^2 = 0.912989$	WSD MAPE = 1.4924%; 8-1-1 MAPE = 1.4925%
MPL baseline	Train log-space $R^2 = 0.824753$	Test log-space $R^2 = 0.805011$

The lightweight progress-aware Tissue variant improves the plain baseline, while MPL provides a more expressive reproduction baseline on the same pkl data.

What we conclude

- Cross-schedule loss prediction is feasible, but the trajectory is highly schedule-sensitive.
- The plain Tissue baseline has a clear early transient error.
- The normalized-progress correction reduces that error and improves test MAPE.
- MPL remains an important reproduced baseline and contrast case.

Final choice: use the progress-aware Tissue variant as the project's own lightweight method, and use MPL as the second reproduced baseline.

Limitations and next steps

- The current improvement is still lightweight and may not capture every schedule change.
- A more expressive schedule-aware term is a natural next step.
- If time allows, the final version can add one more ablation or a stronger schedule encoding.

Thank you.